EDUGENIE – PROJECT DEVELOPMENT PHASES

4. Project Planning Phase

The project planning phase determines how the project will be implemented and divided into manageable activities.

Project Objectives

The main objectives were:

- Develop an AI-powered educational assistant.
- Integrate Generative AI into an educational application.
- Create a simple web interface.
- Develop multiple independent learning modules.
- Connect the frontend with FastAPI APIs.
- Test each feature independently.
- Demonstrate the complete working system.

Development Plan

Phase	Activity
Phase 1	Brainstorming and idea selection
Phase 2	Requirement analysis
Phase 3	System and UI design
Phase 4	Backend development
Phase 5	AI module integration
Phase 6	Frontend development
Phase 7	Testing and debugging
Phase 8	Documentation and demonstration

Technology Planning

The project uses Python-based backend technologies and web technologies for the frontend. Cloud-based AI provides the main generative capabilities while lightweight local processing can be used for selected explanation tasks.

Risk Planning

Possible challenges identified during planning included:

- AI-generated incorrect responses.
- API failures.
- Invalid JSON responses from quiz generation.
- Internet dependency.
- API key security.
- Slow response for some AI operations.
- Compatibility issues between frontend and backend.

Appropriate error handling and testing were planned to reduce these problems.